

Submission & Reviewer Matching

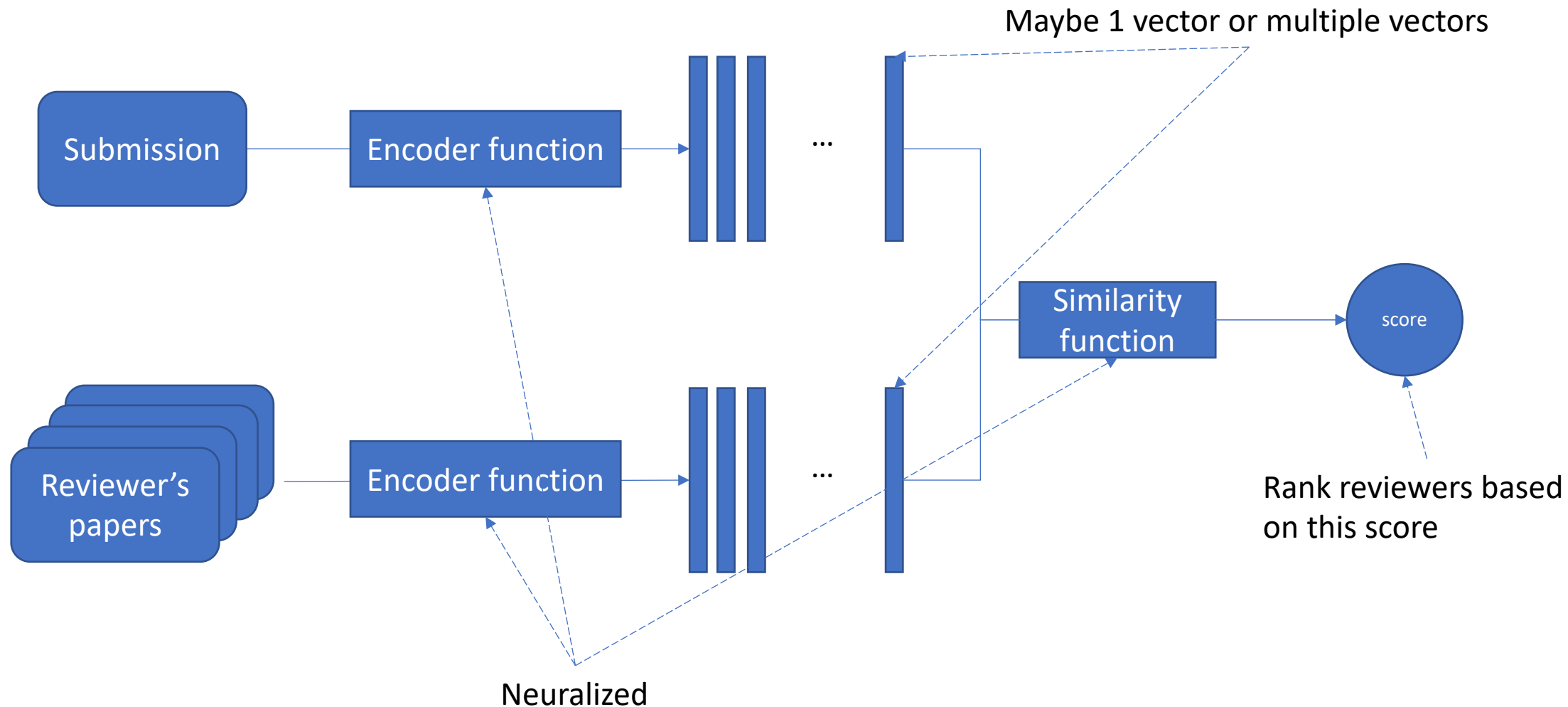
Weekly update from Qiang

07/09/2019

Project goal

1. A useful system that has direct impact on the review processes of academic conferences
2. Scientific contribution on document similarity

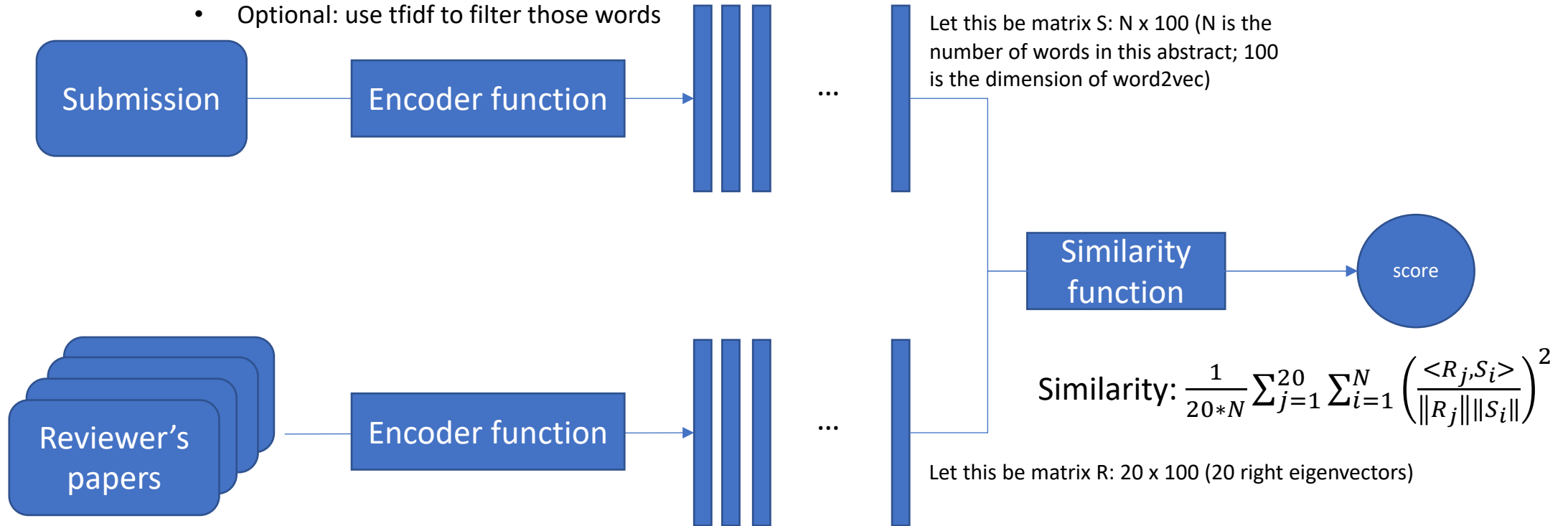
High-level system design



**I didn't discuss conflict-of-interest issues and assignment algorithmic issues.*

What we have now

- Pre-train word2vec on all the papers in our corpus
- Stack all the word vectors from the abstract of the submission
- Optional: use tfidf to filter those words

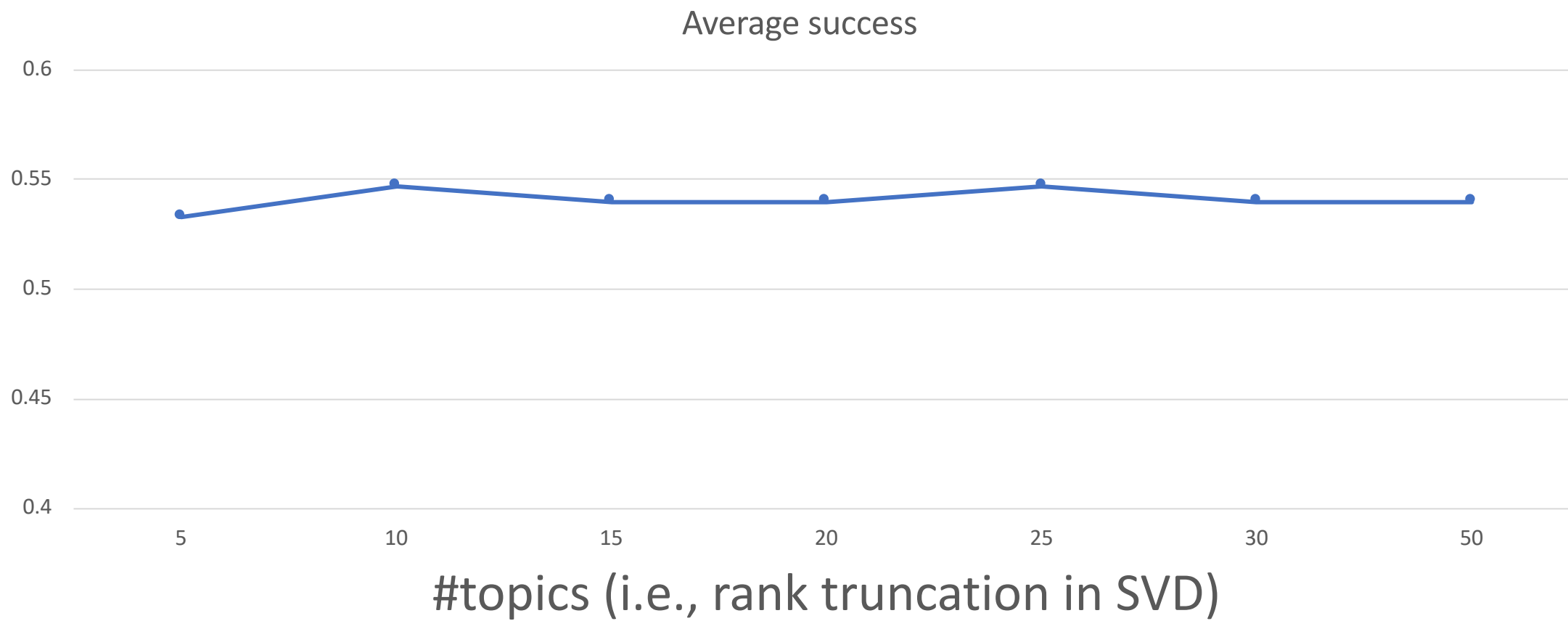


- Pre-define a weight vector for papers in specific years (putting more weight on more recent papers)
- Stack all the word vectors from the abstract of the papers of one particular review
- Perform SVD and obtain the first 20 right eigenvectors
- Optional: use tfidf to filter those words

Recent improvement

- Use matrix multiplication and remove redundancy in calculation
- To match between 356 reviewers and 147 submissions
 - Old system: ~3 hours
 - New system: ~2 mins
 - (the most recent system: ~ 1min)
 - Note that the similarity scores from both old and new systems are exactly the same (which means we reduced the computational time without sacrificing the performance at all).

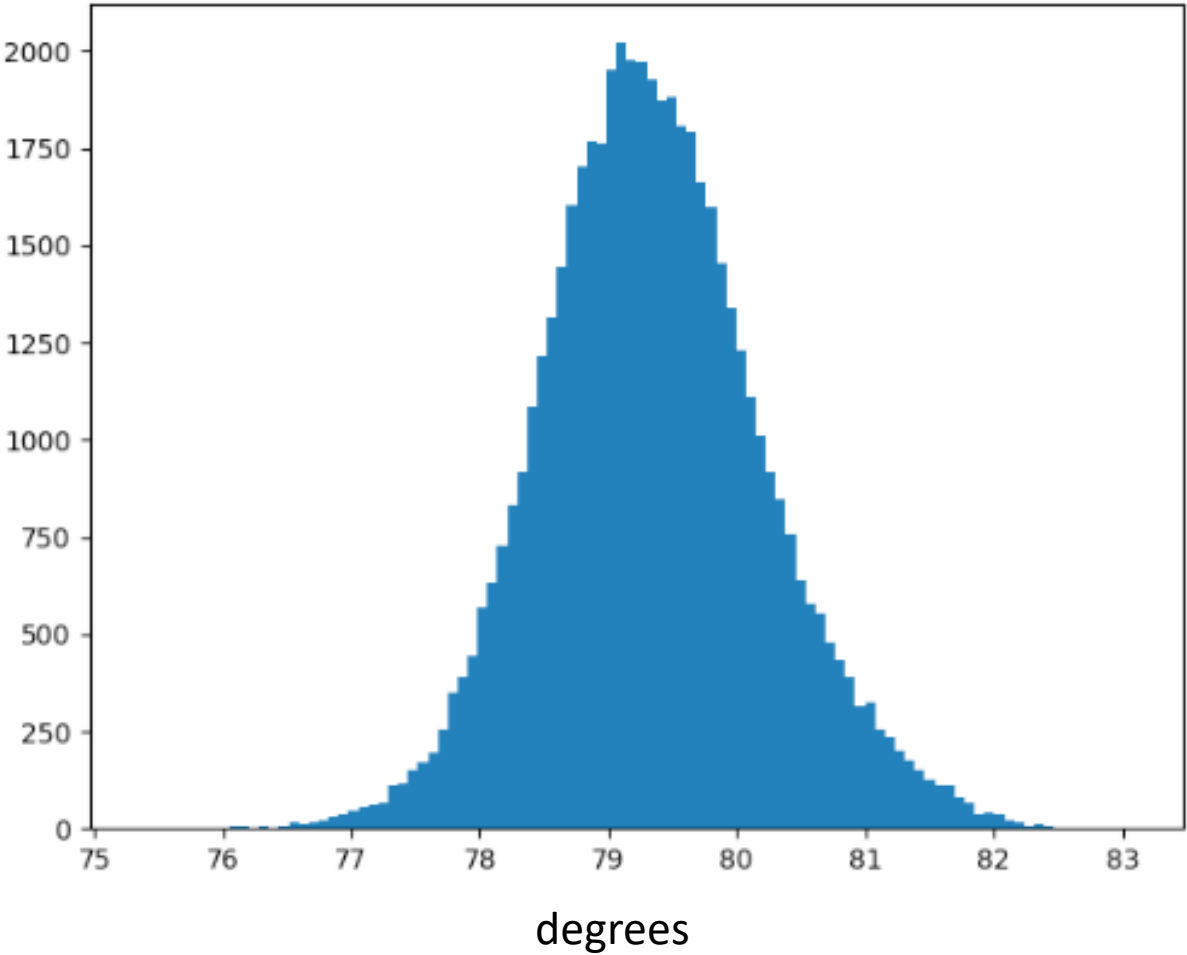
Performance of different #topics



paper reviewer Relevance scores

0	1	0.17703180405188945
0	2	0.18176483430439697
0	3	0.18114263471065978
0	4	0.18468885255109688
0	5	0.17605921817474693
0	6	0.18420585318366145
0	7	0.1880475607444184
0	8	0.20095696727746168
0	9	0.18537059797480732
0	10	0.17969641688689109
0	11	0.17696326359155884
0	12	0.18173867803894658
0	13	0.162947634434677
0	14	0.18215936292728474
0	15	0.16308125749104513
0	16	0.18011719703361245
0	17	0.17156921883988205
0	18	0.170366918236855
0	19	0.16788674385085314
0	20	0.1738794554579193
0	21	0.17875609476160353
0	22	0.18008602487959258
0	23	0.18208989013692525

Histogram of all these scores



Future plan

- Re-organize the code. This is important for future experiments.
- Double check every step. For example,
 - Correct the TFIDF error
 - Keywords extraction
 - SVD: whether SVD is good for this problem
 - Try other similarity metrics
 - Run word2vec or GloVe more carefully
- Run AI2's citation matching system as a baseline

Tentative system

